Electrochemistry

Masterclass Study Notes &
Theoretical Frameworks

Electrochemistry

Potential - tendency to disappear.

Standard electrode potential (E°) - It is the measure of tendency of a rxn to occur.

$$\left.\begin{array}{l} Na \rightarrow Na^+ + e^- \\ K \rightarrow K^+ + e^- \end{array}\right\} \rightarrow \text{Oxidation potential - since losing } e^- \text{ is an ox. rxn.}$$

Notice how Na easily loses an electron to become stable!

$$\left.\begin{array}{l} Cl_2 + 2e^- \rightarrow 2Cl^- \\ F_2 + 2e^- \rightarrow 2F^- \end{array}\right\} \rightarrow \text{Reduction potential - since gain of } e^- \text{ is red. rxn.}$$

Higher ox./red pot. indicates easiness of rxn to occur in forward direction.

$E^\circ_{oxi,\,Na} > E^\circ_{red,\,Cl} \rightarrow$ Na will give e^- to Cl^-

$\downarrow$

Na^+ & Cl^- will get formed.

Galvanic cells :-

→ Electrochem cells.

→ Louis Galvani introduced in 1780. (frog exp)

→ Alexandro Volta generalised it in 1795 by using term "electricity fluid". Also made Voltaic piles.

$$Zn + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s) \; :-$$

$$Zn(s) \xrightarrow{aq.} Zn^{+2}(aq) \; : \; Zn(s) \,/\, Zn^{+2}(aq)$$

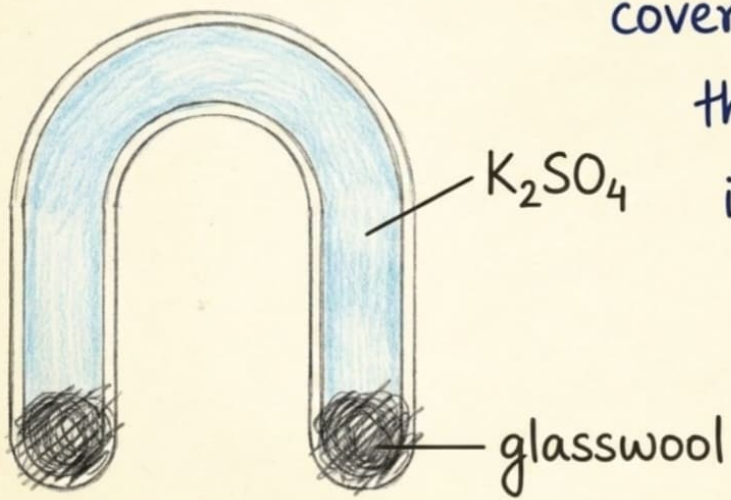
$$Cu^{+2}(aq) \longrightarrow Cu(s) \; : \; Cu^{+2}(aq) \,/\, Cu(s)$$

$$Zn(s) \mid Zn^{+2}(aq) \;\|\; Cu^{+2}(aq) \mid Cu(s)$$

Salt bridge (phase separator)

Salt bridge :-

1. U-shaped pipe containing conc. K_2SO_4 with its ends covered with glasswool so that the K_2SO_4 remains inside the pipe.

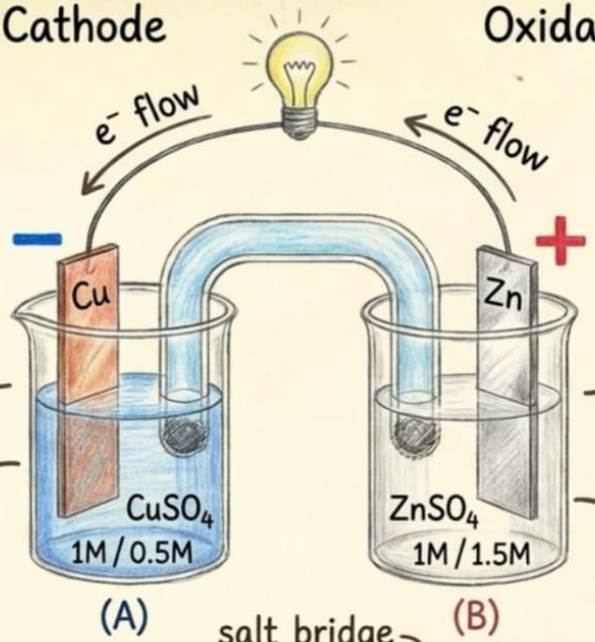


Reduction occurs at Cathode

Oxidation occurs at anode

However, Cu^{+2} ions are getting deposited on Cu plate from soln, ∴ net charge in beaker (A) is -ve.

Since Zn^{+2} ions are getting detached from Zn plate into soln, ∴ net charge in beaker (B) is +ve.



Now, to neutralize, SO_4^{2-} ions start going into (B) through salt bridge by diff. of K_2SO_4, and thus the +vely charged beaker becomes neutralized.

However, the salt bridge becomes +vely charged, for which SO_4^{2-} in (A) goes into salt bridge & neutralises it.

salt bridge

$Zn(s) \mid Zn^{2+}(aq)\,(0.1M) \parallel Cu(s) \mid Cu^{2+}(aq)\,(0.1M)$

oxi.

red.

phase boundary separator

Nernst eq. :-

$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$\Delta G^\circ = -nFdE^\circ_{cell}$$

$$\therefore \; -nFdE_{cell} = \Delta G^\circ + RT \ln Q$$

$$\text{or} \quad E_{cell} = E^\circ + \frac{RT \ln Q}{-nFd}$$

$$\boxed{\therefore \; E_{cell} = E^\circ - \frac{0.0592}{n} \log_{10} Q \quad \text{at } 25^\circ C.}$$

$$\boxed{pH = \frac{E_{cell} - E_{ref}}{0.0592}}$$

$$0.0592\ pH = E_{H_2/H^+,\,oxi}$$

$$\boxed{E_{H^+/H_2} = -0.0592\ pH}$$

At equilibrium, $E_{cell} = 0$,

$$\therefore E^\circ_{cell} = \frac{0.0592}{n} \log K$$

$$\Delta G^\circ = -2.303\, RT \log K_c$$

$$\therefore \text{At equilibrium, } -nFdE^\circ_{cell} = -2.303\, RT \log K_c$$

$$\text{or} \quad nFdE^\circ_{cell} = 5705.8 \log K_c$$

5705.8 comes from multiplying 2.303 × R(8.314) × T(298K)!

Faraday's law of electrolysis :-

1. Charge of 1 mol of e^- is 1Fd = 96500 C/mol.
2. equivalent mass = $\frac{\text{mass of element}}{\text{charge on it}}$
3. Amt of electricity to be passed in (Fd) = No. of charged units present on an atom.

e.g. Ag^+ needs 1Fd to be passed for getting 1 mol of Ag to be deposited.

$\therefore M = EQ$

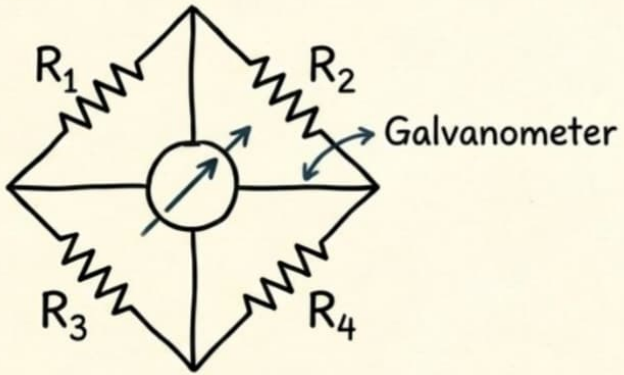
where M is the mass, Q is in Fd & E is equivalent wt.

∴ For 124g of Au, 1Fd is needed to pass get 1 mol of Au.

$$\frac{m_1}{m_2} = \frac{E_1}{E_2}$$

Measurement of conductance :-

* Wheatstone bridge :-



$$\frac{R_1}{R_3} = \frac{R_2}{R_4}$$

Incase of Balanced bridge (i.e, when there is no charge moving through conductors).

Invented by Samuel Hunter Christie & improved by Charles Wheatstone.

Electrolytic Solutions – conduction of electricity :-

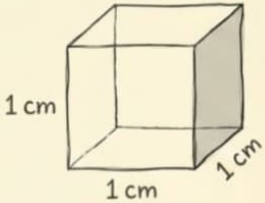
* Arrhenius theory of electrolytes :-

α, also called as degree of diss/ioni. refers to the no. of molecules diss or ionised after adding into solvent.

* Electrolytic conductance :-

Resistance is the response shown by 2 opposing charges to each other due to which they come closer and move away while being in a solvent.

Resistivity is the resistance among 2 opp charges in $1\,cm^3$ volm cube.



$$\therefore R = \rho \frac{\ell}{A} \quad (\Omega cm)$$

Specific conductance (conductivity k or κ) is the conductance in 1 cm^3 volume & is also the reciprocal of ρ.

$$\therefore K = \frac{1}{\rho} = \frac{1}{R} \times \frac{\ell}{A} \quad (\Omega^{-1}cm^{-1} \text{ or } Scm^{-1})$$

It does not depend upon the volume of solute rather it depends upon the max no. of ions which can be ionised or diss from the molecules of solute.

* <u>Molar conductivity & equivalent conductivity</u> :-

Molar conductivity is the conductn of 1 mole of solute. (Λ_m)

Equivalent conduct. is the conduct^n of 1 eq of solute (Λ_{eq}).

$$\Lambda_m = \Lambda_{(+)} \text{ or } (-) \text{ charge} \times \Lambda_{eq}.$$

$$\Lambda_m = \kappa \times V \quad \& \quad \Lambda_{eq} = \kappa \times V$$

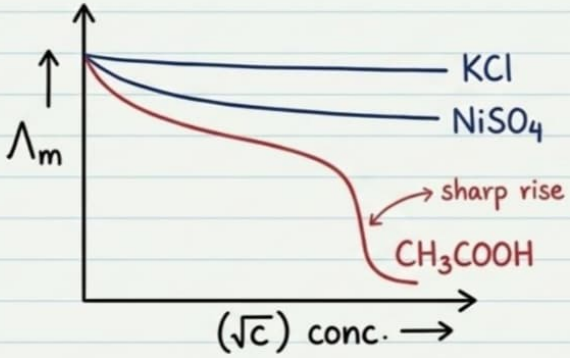
e.g. $CaCl_2 \longrightarrow Ca^{+2} + 2Cl^-$

Since, Ca^{2+} has 2 +ve charges present,
$\therefore$ It has 2 equivalents

or equivalent mass $= \dfrac{\text{molar mass}}{2} = 55.5g$

$$\therefore \Lambda_{m,\, Ca^{+2}} = 2 \times \Lambda_{eq,\, Ca^{+2}}$$

Kohlausch's law :-



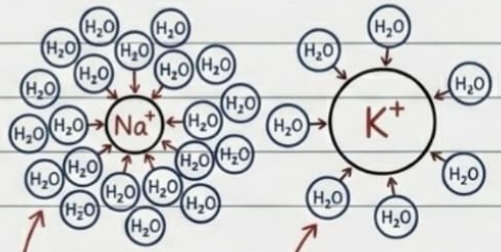
$$\Lambda_m = \Lambda^\circ - K\sqrt{c}$$

The dilution has been done to this much extent that the conc. of $NiSO_4$ & KCl becomes very low as almost 100% of diss or ionisation has been occurred. However, as seen in graph at ∞ dil., KCl will be more conducting than $NiSO_4$.

However, in case of CH_3COOH, it is a weaker acid thus it ionizes to a least amount, however as dil. inc, we can see a sharp rise in its graph.

Masterclass Notebook :-

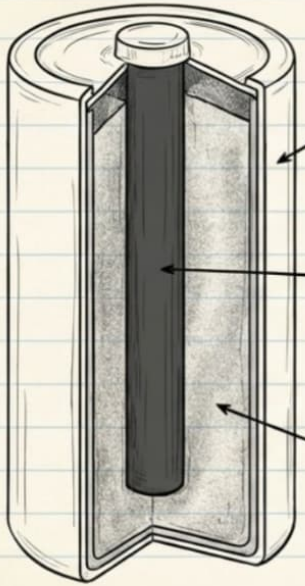
Conductivity of $Na^+ < K^+$, due
(inn to its small size and thus
more charge ρ, $\therefore$ it has more
$\therefore$ it has more
$\sum$ it as due to which
more H_2O molecules will get
attached to it & it thus
becomes heavier,
$\therefore$ it will move slower.



Smaller ion = greater charge density = more hydration = slower movement!

$\therefore$ Its mobility is lesser.

2.6.1 Primary Batteries



At diff - - 1.5V

Commercial dry cell cylinder (Fig 2.8)

reaction occurs only once and after use over a period of time battery becomes dead and cannot be reused again.

dry cell (known as Leclanche cell after its discoverer)

consists of a zinc container that also acts as anode and the cathode is a carbon (graphite) rod surrounded by powdered manganese dioxide and carbon

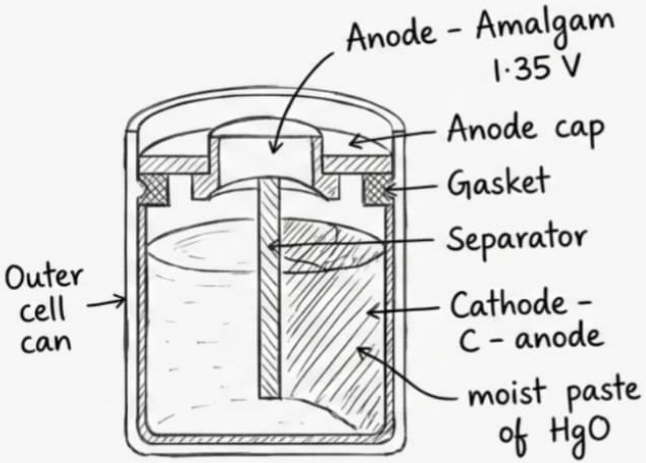
moist paste of ammonium chloride (NH_4Cl) and zinc chloride ($ZnCl_2$)

Anode: $Zn(s) \rightarrow Zn^{2+} + 2e^-$

Cathode: $MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$

manganese is reduced from the + 4 oxidation state to the +3 state. Ammonia produced in the reaction forms a complex with Zn^{2+} to give $[Zn(NH_3)_4]^{2+}$

The cell has a potential of nearly 1.5 V.



Mercury cell (Fig 2·9)

Mercury cell, suitable for low current devices like hearing aids, watches, etc. devices like hearing aids, watches, etc. consists of zinc - mercury amalgam as anode and a paste of HgO and carbon as the cathode. The electrolyte is a paste of KOH and ZnO.

Anode: $Zn(Hg) + 2OH^- \longrightarrow ZnO(s) + H_2O + 2e^-$

Cathode: $HgO + H_2O + 2e^- \longrightarrow Hg(\ell) + 2OH^-$

The overall reaction is represented by

$$Zn(Hg) + HgO(s) \longrightarrow ZnO(s) + Hg(\ell)$$

The cell potential is approximately 1·35 V and remains constant during its life as the overall reaction does not involve any ion in solution whose concentration can change during its life time.

2.6.2 Secondary Batteries

recharged by passing current through it in the opposite direction so that it can be used again. A lead storage battery commonly used in automobiles and invertors. It consists of a lead anode and a grid of lead packed with lead dioxide (PbO_2) as cathode. A 38% solution of sulphuric acid is used as an electrolyte.

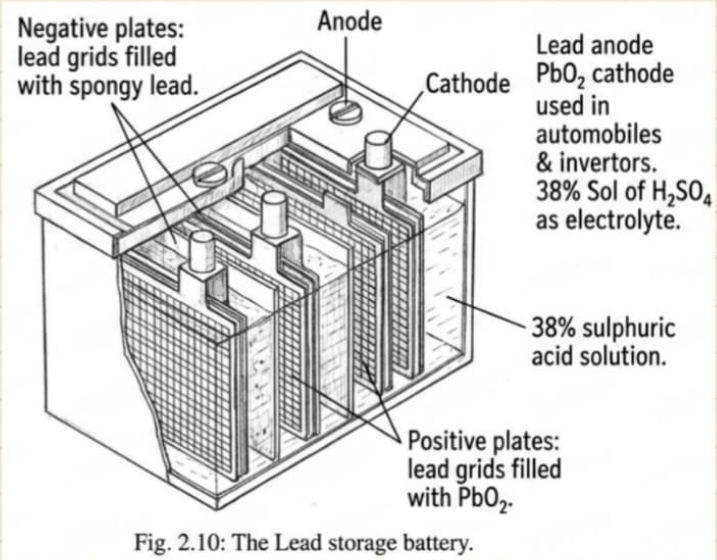


Fig. 2.10: The Lead storage battery.

Anode: $Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$

Cathode: $PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$

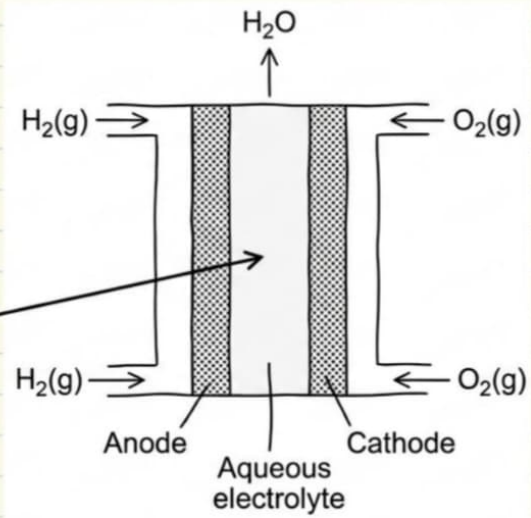
Cell reactions during discharge.

2.7 Fuel Cells

Galvanic cells that are designed to convert the energy of combustion of fuels like hydrogen, hydrogen, methane, methanol, etc. directly into electrical energy are called fuel cells.

Used for provid electrical power to Apollo space prog.

Pt & Pd were used as catalysts.



H_2/O_2 fuel cell chamber (Fig 2.12)

fuel cells uses the reaction of hydrogen with oxygen to form water

The water vapours produced during the reaction were condensed and added to the drinking water supply for the astronauts.

eff of fuel cells = 70%.
eff of thermal plants = 40%.

Cathode: $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$

Anode: $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$

Overall reaction being: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

2.8 Corrosion

Corrosion slowly coats the surfaces of metallic objects with oxides or other salts of the metal. rusting of iron, tarnishing of silver, development of green coating on copper and bronze are some of the examples of corrosion. We lose crores of rupees every year on account of corrosion.

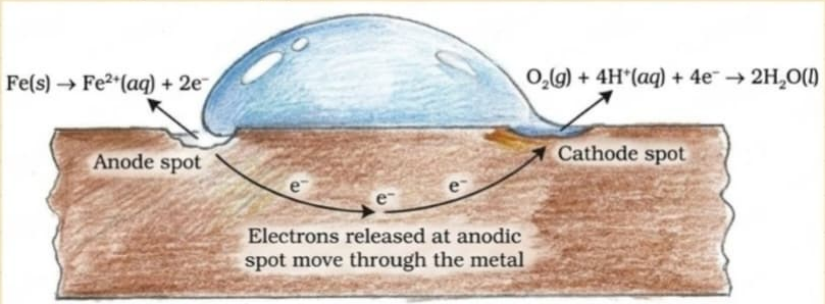


Fig. 2.13: *Corrosion of iron in atmosphere*

Anode: $2Fe(s) \rightarrow 2Fe^{2+} + 4e^-$ $\quad E°(Fe^{2+}/Fe) = -0.44$ V

Electrons released at anodic spot move through the metal and go to another spot on the metal and reduce oxygen in the presence of H^+… This spot behaves as cathode with the reaction

Cathode: $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$ $\quad E°(H^+/O_2/H_2O) = 1.23$ V

The overall reaction being: $2Fe(s) + O_2(g) + 4H^+(aq) \rightarrow 2Fe^{2+}(aq) + 2H_2O(l)$

$E°(cell) = 1.67$ V

The ferrous ions are further oxidised by atmospheric oxygen to ferric ions which come out as rust in the form of hydrated ferric oxide ($Fe_2O_3 \cdot xH_2O$) and with further production of hydrogen ions.

Prevention of Corrosion

- ☐ prevent the surface of the metallic object to come in contact with atmosphere.

  (bisphenol is used to avoid rusting.)

- ☐ cover the surface by other metals (Sn, Zn, etc.) that are inert or react to save the object.

- ☐ An electrochemical method is to provide a sacrificial electrode of another metal (like Mg, Zn, etc.) which corrodes itself but saves the object.

  ↳ Sacrificial electrodes = the ultimate electrochemical defense!